

Welcome to CS 401/MCS 401

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- content
- expectations

2 The Stable Matching Problem

- problem statement
- stable matching
- the Gale-Shapley algorithm
- analyzing the algorithm
- termination and return of stable matching
- unique outcome

CS 401/MCS 401 Lecture 1
Computer Algorithms I
Jan Verschelde, 15 June 2026

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Catalog Description

CS 401/MCS 401 – Computer Algorithms I

Design and analysis of computer algorithms.

Divide-and-conquer, dynamic programming, greedy method, backtracking.

Algorithms for sorting, searching, graph computations, pattern matching, NP-complete problems.

Prerequisite(s): Grade of C or better in MCS 360;
or Grade of C or better in CS 251.

Definition (0)

An *algorithm* is an *ordered*₍₁₎ set of *unambiguous*₍₂₎, *executable*₍₃₎ steps that define a *terminating*₍₄₎ process.

Source: *Computer Science, an Overview* by J. Glenn Brookshear.

Content and Organization of the Course

The order of topics covered in the course follows
Jon Kleinberg and Éva Tardos: *Algorithm Design*, Pearson, 2006.

Chapters we will cover:

- 1 the Gale-Shapley algorithm
- 2 basics of algorithm analysis
- 3 introduction to graphs
- 4 greedy algorithms
- 5 divide and conquer
- 6 dynamic programming
- 7 network flow
- 8 NP and computational intractability

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Purpose of the Course

The goals of CS 401/MCS 401 are

- 1 solve discrete problems in computer science;
- 2 design and analysis of algorithms; and
- 3 reason mathematically about correctness and execution costs.

There are two exams:

- 1 The midterm exam is worth 100 points.
- 2 The final exam counts for 200 points.

Homework: 100 points, collected every week, except exam weeks.

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the stable matching problem

Consider the job market:

- Employers have job openings to fill.
- Applicants apply to the job openings.

After a round of interviews:

- Every employer has ranked applicants based on qualifications.
- Every applicant has ranked employers based on preferences.

After a round of job offers and acceptances, we ask the question.

Is there a *stable* matching?

- Every employer prefers every one of its accepted applicants, over every other applicant who was not received an offer.
- Every applicant prefers the current employer over every other employer who has made an offer.

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the marriage problem

In the stable matching problem,

- while every applicant can accept only one job,
- an employer may have many jobs, and
- there may not be a job for every applicant.

To eliminate complications caused by asymmetries:

- n applicants apply to n employers, and
- each employer accepts only one single applicant.

An alternative formulation is the stable marriage problem:

In the marriage problem we want to arrange n marriages for n men and n women who want to get married.

perfect matchings

Consider a set of n employers: $E = \{e_1, e_2, \dots, e_n\}$
and a set $A = \{a_1, a_2, \dots, a_n\}$ of n applicants.

Denote by $E \times A = \{ (e, a) \mid e \in E, a \in A \}$
the set of all possible pairs of employers and applicants.

Definition (1)

A subset S of $E \times A$ is a *matching*
if each $e \in E$ and each $a \in A$ appears at most once in S .

Definition (2)

A matching S is a *perfect matching*
if each $e \in E$ and each $a \in A$ appears exactly once in S .

preferences and stable matchings

Every employer $e \in E$ ranks every applicant $a \in A$, no ties allowed.
Every applicant $a \in A$ ranks every employer $e \in E$, no ties allowed.

If e ranks a higher than a' , then we say: e *prefers* a to a' .

If a ranks e higher than e' , then we say: a *prefers* e to e' .

Definition (3)

Two pairs (e, a) and (e', a') have the *instability* (e, a') if

- 1 e prefers a' to a ; and
- 2 a' prefers e to e' .

For a matching S , if $(e, a), (e', a') \in S$ have the instability (e, a') , then (e, a') is an instability with respect to S .

Definition (4)

A matching S is *stable* if S is perfect and there are no instabilities with respect to S .

an input to our problem

Consider 4 employers $\{1, 2, 3, 4\}$ and 4 applicants $\{a, b, c, d\}$.

preferences for employers

1 : a c b d

2 : b a d c

3 : a c b d

4 : d a c b

preferences for applicants

a : 2 3 4 1

b : 1 4 2 3

c : 1 3 4 2

d : 2 1 3 4

Read the first line as

- employer 1 prefers a to c , prefers c to b , prefers b to d ,
- applicant a prefers 2 to 3, prefers 3 to 4, prefers 4 to 1.

Our problem: find a perfect stable matching.

a very specific input to our problem

- Instead of employers, consider 4 mentors $\{1, 2, 3, 4\}$, ranked according to reputation: $1 > 2 > 3 > 4$.
- The applicants are apprentices $\{a, b, c, d\}$, ranked according to qualifications $a > b > c > d$.

Naturally, any mentor wants the best qualified apprentice, and any apprentice wants the mentor with the best reputation.

preferences for mentors

1	:	a	b	c	d
2	:	a	b	c	d
3	:	a	b	c	d
4	:	a	b	c	d

preferences for apprentices

a	:	1	2	3	4
b	:	1	2	3	4
c	:	1	2	3	4
d	:	1	2	3	4

Exercise 1: What is a (or the) perfect stable matching for this problem?
Justify your answer. Prove the correctness of your answer.

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setup of the algorithm

Denote: E is the set of employers $e \in E$, $\#E = n$;
 A is the set of applicants $a \in A$, $\#A = n$;
 S is the set of committed pairs $(e, a) \in S \subset E \times A$.

Every employer and every applicant has a preference list with no ties.

Queries: $\text{isfree}(e)$: there is no $a \in A$: $(e, a) \in S$;
 $\text{isfree}(a)$: there is no $e \in E$: $(e, a) \in S$;
 $\text{hasoffered}(e, a)$: e has made a job offer to a .

Initially: for all $e \in E$: $\text{isfree}(e)$ is true,
for all $a \in A$: $\text{isfree}(a)$ is true, and
 $\text{hasoffered}(e, a)$ is false.
 $S = \emptyset$.

the Gale-Shapley algorithm

Repeat the following statements:

- 1 Let $e \in E$: $\text{isfree}(e)$ and
there is an $a \in A$: $\text{hasoffered}(e, a)$ is false.

If no such e exists, then leave the repeat loop, return S .

- 2 Let $a \in A$: a is the highest ranked in the preference list of e
for whom $\text{hasoffered}(e, a)$ is false.

- 3 $\text{hasoffered}(e, a) := \text{true}$

If $\text{isfree}(a)$ then

$S := S \cup (e, a)$ (e and a are committed)

$\text{isfree}(e) := \text{false}; \text{isfree}(a) := \text{false}$

else

there is an $e' \in E$: $(e', a) \in S$ (a is committed to e')

if a prefers e to e' then

$S := S \setminus (e', a)$ (e' and a are no longer committed)

$\text{isfree}(e') := \text{true}$ (e' becomes free)

$S := S \cup (e, a)$ (e and a are committed)

running the Gale-Shapley algorithm

Consider 4 employers $\{1, 2, 3, 4\}$ and 4 applicants $\{a, b, c, d\}$.

preferences for employers

1 : a c b d

2 : b a d c

3 : a c b d

4 : d a c b

preferences for applicants

a : 2 3 4 1

b : 1 4 2 3

c : 1 3 4 2

d : 2 1 3 4

Read the first line as

- employer 1 prefers a to c , prefers c to b , prefers b to d ,
- applicant a prefers 2 to 3, prefers 3 to 4, prefers 4 to 1.

Exercise 2: Run the algorithm, show the evolution of S in each step.

Is the returned set of pairs a perfect matching?

Verify whether it is stable.

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analyzing the algorithm – viewpoint of an applicant

The Gale-Shapley algorithm looks simple, but it is not obvious that the returned set of pairs is a stable matching.

Consider the viewpoint of an applicant during the algorithm:

- If free, an applicant accepts the first offer.
- If committed, then an additional offer is accepted only if the employer is ranked higher in the list of preferences.
- After the first offer, an applicant stays committed.

An applicant ends up committed to the highest ranked employer, highest ranked of all employers who made a job offer to the applicant.

analyzing the algorithm – viewpoint of an employer

Consider the viewpoint of an employer during the algorithm:

- An employer remains free until an offer made to the highest ranked applicant (who has not been offered to) is accepted.
- Once committed, an employer may become free again.

During the running of the algorithm, the ranking of applicants in the preference list of the employer who offers gets worse.

Recall the viewpoint of an applicant, who gets committed to better employers.

a third exercise

adapted from exercise 1 of the textbook

Exercise 3:

In every instance of the stable matching problem, there is a stable matching containing a pair (e, a) , such that

- 1 e is ranked first in the preference list of a ; and
- 2 a is ranked first in the preference list of e .

Is this true or false?

- If true, give a short explanation.
- If false, give a counterexample.

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termination and the number of steps

count the offers

Theorem (5, termination)

For n employers and n applicants, the Gale-Shapley algorithm terminates after at most n^2 iterations in the loop.

Proof. We measure the progress of the algorithm by

$$\mathcal{P}(t) := \#\{ (e, a) \mid \text{hasoffered}(e, a) \}$$

at iteration t in the loop. We have $\mathcal{P}(t) < \mathcal{P}(t+1)$ because

- In each iteration, we execute $\text{hasoffered}(e, a) := \text{true}$,
- except when we leave the loop at the first instruction.

The total number of pairs is $n \times n$, which gives an upper bound on the total number of offers.

$\Rightarrow \mathcal{P}(t)$ can increase at most n^2 times.

Q.E.D.

no employer falls off the end of the preference list

Lemma (6)

If an employer is free at some point in the execution of the algorithm, then there is an applicant to whom the employer has not yet offered to.

Proof. (by contradiction)

Assume an employer is free and has already offered to every applicant.

Recall the analysis from the viewpoint of an applicant:

- once received an offer, an applicant is no longer free;
- either the applicant stays in the current commitment, or becomes committed to the employer who makes an offer to the applicant if that employer is higher in the ranking of the applicant.

The assumption implies then that all n applicants are committed.

As one applicant is paired with only one employer, all n employers are committed.

This leads to a contradiction, so the assumption is false.

Q.E.D.

we have a perfect matching

Lemma (7)

The set returned at termination is a perfect matching.

Proof. First, we verify that the set of pairs is a matching via the analysis from the viewpoint of an applicant:

- An applicant is either free, or committed to only one employer.
- So, every applicant appears only once in the set.

An employer who is committed does not offer.

- An employer is either free or committed to one applicant.
- So, every employer appears only once in the set.

By the previous two sentences, starting with “So,” the set returned is a matching.

still have to show the matching is perfect

Second, we show the set is a perfect matching by contradiction.

Assume the algorithm terminates with a free employer.

By the previous Lemma (6), no employer falls off the end of the preference list: there is an applicant to whom that free employer has not yet offered.

But the algorithm terminates only if there is no free employer who has not yet offered to every applicant.

This contradicts the assumption,
therefore, we have a perfect matching.

Q.E.D.

Theorem (8, correctness)

The set S returned by the Gale-Shapley algorithm is a stable matching.

Proof. By the previous Lemma (7), we have a perfect matching.

We prove stability by contradiction.

Assume $(e, a), (e', a') \in S$ have the instability (e, a') :

e prefers a' to a and a' prefers e to e' .

The last offer of e was to a , because of $(e, a) \in S$.

Did e offer to a' ?

We have to examine both possibilities ...

correctness proof continued

We assumed an instability: e prefers a' to a and a' prefers e to e' .

- If e did not offer to a' , then a is higher on the preference list of e , contradicting the assumption that e prefers a' to a .
- If e did offer to a' , then there is some e'' :
 - ▶ either a' was committed to e'' , ranked higher than e , and therefore a' rejected the offer of e ;
 - ▶ or a' received an offer from e'' , ranked higher than e , and therefore a' broke off the commitment to e .

If $e' = e''$, then we have that a' prefers $e'' = e'$ over e , contradicting the assumed instability.

Otherwise, if $e' \neq e''$, then a' prefers e' to e'' , a' prefers e'' to e , and by transitivity, a' prefers e' to e .

This contradicts the assumed instability and therefore, S is a stable matching.

Q.E.D.

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running the Gale-Shapley algorithm again

Consider 4 employers $\{1, 2, 3, 4\}$ and 4 applicants $\{a, b, c, d\}$.

preferences for employers

1 : a c b d

2 : b a d c

3 : a c b d

4 : d a c b

preferences for applicants

a : 2 3 4 1

b : 1 4 2 3

c : 1 3 4 2

d : 2 1 3 4

Read the first line as

- employer 1 prefers a to c , prefers c to b , prefers b to d ,
- applicant a prefers 2 to 3, prefers 3 to 4, prefers 4 to 1.

Exercise 4: Run the algorithm, swapping employers with applicants: applicants offer, employers accept or reject. Show the evolution of S .

Compare S with the run of the algorithm in Exercise 2.

Does the order of the proposals matter?

Consider the case where each employer has a different first choice:

- The outcome does not depend on the order of proposals; and
- the outcome is unique.

In this case, the preferences of applicants do not matter.

The first step in the loop starts with “Let $e \in E$,”
not requiring an order on the set E of employers.

Can a different order on E lead to a different outcome?

characterization of the unique outcome

Definition (9)

An applicant a is a *valid partner* of an employer e if there is a stable matching that contains (e, a) .

Definition (10)

An applicant a is the *best valid partner* of e (denoted by $\text{best}(e)$) if

- 1 a is a valid partner of e ; and
- 2 no applicant whom e ranks higher than a is a valid partner of e .

Denote $S^* = \{ (e, \text{best}(e)) \mid e \in E \}$.

Theorem (11)

Every execution of the Gale-Shapley algorithm results in S^ .*

Every employer is paired to the best valid partner.

proof by contradiction

Theorem (11)

Every execution of the Gale-Shapley algorithm results in S^ .*

Assume some run results in an S : $(e, a) \in S$ and $a \neq \text{best}(e)$.

Since employers offer in decreasing order, e was rejected by $\text{best}(e)$:

- either $\text{best}(e)$ was committed to e' ranked higher than e ;
- or $\text{best}(e)$ was committed to e , but broke off the commitment when e' ranked higher than e made a job offer.

In both cases, $\text{best}(e)$ prefers e' to e .

Since $\text{best}(e)$ is a valid partner of e , there exists a stable matching S' : $(e, \text{best}(e)) \in S'$. Consider e' , the employer committed to $\text{best}(e)$ in S . In S' , $(e', a') \in S'$, $a' \neq \text{best}(e)$.

The assumption will lead to an instability in S' , contradicting that S' is a stable matching.

proof by contradiction continued

- 1 $\text{best}(e)$ prefers e' to e , as implied by our assumption.
- 2 $(e, \text{best}(e)) \in S'$, for some S' , by definition of $\text{best}(e)$.
- 3 $(e', a') \in S'$, $a' \neq \text{best}(e)$.

In the run which resulted in S , $(e, a) \in S$ and $a \neq \text{best}(e)$:

- Since the rejection of e by $\text{best}(e)$ was the first rejection of an employer by a valid partner, e' had not been rejected by a valid partner when e' became committed to $\text{best}(e)$.
- Since e' offered in decreasing order of preference and since a' is a valid partner of e' , e' prefers $\text{best}(e)$ to a' .

If e' prefers $\text{best}(e)$ to a' and $\text{best}(e)$ prefers e' to e , then (e', a') is an instability with respect to S' .

The instability contradicts that S' is a stable matching.

The assumption is false. Therefore, $S = S^*$.

Q.E.D.

the outcome evaluated for applicants

Definition (12)

An employer e is *a valid partner* of an applicant a if there is a stable matching that contains (e, a) .

Definition (13)

An employer e is *the worst valid partner* of an applicant a if

- 1 e is a valid partner of a ; and
- 2 no employer whom a ranks lower than e is a valid partner of a .

Corollary (14)

In the result S^ of the Gale-Shapley algorithm, each applicant is paired with the worst valid partner.*

Every employer is paired to the best valid partner.

proof by contradiction

Corollary (14)

In the result S^ of the Gale-Shapley algorithm, each applicant is paired with the worst valid partner.*

Proof. Assume, for $(e, a) \in S^*$, e is not the worst valid partner of a .

Then there is a stable matching S' , $(e', a) \in S'$, a prefers e to e' , by e not being the worst valid partner of a .

In S' , $(e, a') \in S'$, $a' \neq a$, and by the previous Theorem (11): $a = \text{best}(e)$, e prefers a to a' .

Then (e, a') is an instability in S' , contradicting that S' is stable.

Therefore, e is the worst valid partner of a .

Q.E.D.

hospitals and medical students

adapted from exercise 4 on pages 23-24 of the textbook

A version of the Gale-Shapley algorithm ran already before 1962.

Exercise 5:

Consider assigning n medical students to m hospitals, $n > m$.

- Each hospital has a ranking of students, in order of preference; and
- each student has a ranking of hospitals, in order of preference.

Let us assume there are no ties among the preferences.

- Hospitals can take more than one student.
- In general, there is a surplus of medical students.

Show that a version of the Gale-Shapley algorithm finds a stable perfect matching between the students and hospitals.

five representative problems

a very short overview

As a reading assignment, consider five representative problems.

① *Interval Scheduling*

Schedule requests, given by start and finish times, for a resource.

② *Weighted Interval Scheduling*

Each request has a value or a weight.

③ *Bipartite Matching*

Each vertex in a bipartite graph belongs to one of two sets.

Vertices in one set are not connected.

We want to find a matching of the largest size.

④ *Independent Set*

Given a graph, a set of nodes is independent if there is no edge between any two nodes in the set.

⑤ *Competitive Facility Location*

Design a winning strategy in a two player game.